HUMAN SPACEFLIGHT in general—to say nothing of expeditions to Mars—would be much more readily supportable if, as in the fifteenth-century arguments of Columbus and Henry the Navigator, there were a profit lure.* Some arguments have been advanced. The high vacuum or low gravity or intense radiation environment of near-Earth space might be utilized, it is said, for commercial benefit. All such proposals must be challenged by this question: Could comparable or better products be manufactured down here on Earth if the development money made available were comparable to what is being poured into the space program? Judging by how little money corporations have been willing to invest in such technology—apart from the entities building the rockets and spacecraft themselves—the prospects, at least at present, seem to be not very high.

The notion that rare materials might be available elsewhere is tempered by the fact that freightage is high. There may, for all we know, be oceans of petroleum on Titan, but transporting it to Earth will be expensive. Platinum-group metals may be abundant in certain asteroids. If we could move these asteroids into orbit around the Earth, perhaps we could conveniently mine them. But at least for the foreseeable future this seems dangerously imprudent, as I describe later in this book.

In his classic science fiction novel *The Man Who Sold the Moon*, Robert Heinlein imagined the profit motive as the key to space travel. He hadn't foreseen that the Cold War would sell the Moon. But he did recognize that an honest profit argument would be difficult to come by. Heinlein envisioned, therefore, a scam in which the lunar surface was salted with diamonds so later explorers could breathlessly discover them and initiate a diamond rush. We've since returned samples from the Moon, though, and there is not a hint of commercially interesting diamonds there.

However, Kiyoshi Kuramoto and Takafumi Matsui of the University of Tokyo have studied how the central iron cores of Earth, Venus, and Mars formed, and find that the Martian mantle (between crust and core) should be rich in carbon—richer than that